JANANI N

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering Electronics and Communication Engineering student with hands-on experience in PCB design, circuit development, and hardware prototyping**.** Familiar with KiCad, schematic design, PCB layout, component selection, and basic circuit analysis**,** with practical experience in Arduino and sensor-based projects. Possess basic knowledge of C and Embedded C and seeking an entry-level PCB Design Engineer role in hardware development and electronics design.

SKILLS SUMMARY

- PCB Design: KiCad, Schematic Capture, PCB Layout & Routing
- Hardware: Component Selection, Footprint Assignment, Hardware Prototyping
- Embedded: Arduino UNO, ESP8266/ESP32, Sensor Interfacing
- Programming: Basic C & Embedded C
- Tools: KiCad, Arduino IDE

PROJECTS

Adjustable Power Supply using LM317

- Designed and implemented an adjustable voltage regulator PCB using LM317**,** providing an output range of approximately 1.25V–12V**.**
- Created the schematic and PCB layout using KiCad, focusing on proper component placement, routing, and decoupling for stable operation.
- Generated the 3D PCB model and gained practical experience in PCB design, component selection, power supply fundamentals, and hardware prototyping.

LED Blinker Circuit using 555 Timer IC

- Designed an LED Blinker circuit using the 555 Timer IC in astable mode to generate a continuous blinking signal.
- Developed the schematic, assigned footprints and designed the PCB layout using KiCad with proper component placement and routing.
- Integrated Gained practical experience in basic circuit design, PCB routing, and converting electronic schematics into a functional PCB layout

Gas Detection System

- Designed the schematic and PCB layout for an MQ-6 gas detection system using KiCad
- Integrated Arduino Uno, MQ-6 gas sensor, LED, buzzer, and resistors for gas detection and alert indication.
- Implemented PCB routing, vias, and GND copper fill, gaining practical experience in PCB design and analog sensor interfacing.

INTERNSHIP

Internship | *Offline | 15 days*

- Nxtlogic software solution

EDUCATION

B.E., Electronics and Communication Engineering — CGPA: 7.6 | *Kongunadu College of Engineering and Technology, Namakkal, India| Sept 2023 - June 2027*

Higher Secondary Certificate (HSC) — 77% | *Sri Ragavendra Matric Hr. Sec. School, Salem, India | May 2022 - March 2023*

CERTIFICATIONS

- Value added course in PLC
- Value added course in Embedded System with AI Automation
- Industrial Internet of Things - NPTEL

AREA OF INTEREST

- PCB Design & Layout
- Schematic Design & Circuit Development
- Hardware Design & Prototyping
- Electronic Circuit Analysis
- Component Selection & Interfacing
- Embedded Hardware & IoT

Extracurricular Activities

- Familiar with video editing, photo editing, and graphic designing using DaVinci Resolve, Canva, and Photopea.
- Actively developing skills in Embedded Systems and PCB Design through hands-on projects.